Bounding Fourier growth of $\mathbb{F}_2$ polynomials

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Plan for this talk

- Introduction to $\mathbb{F}_2$ polynomials
- An additivity approach to bounding Fourier growth

Recapping the Fourier coefficient

- In the real setting: have the *Fourier transform*

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{i\,\omega x}\, \mathrm{d}\omega$$

  which measures the intensity of a frequency $e^{i\omega\cdot}$ in a function.
  - This can be treated as an inner product $\langle f, e^{i\omega\cdot}\rangle$
  - Used in quantum mechanics, signal processing, nonlinear systems
- The *Fourier coefficient* $\hat{f}(S) = \mathbb{E}_{x\in\{-1,1\}^n}[f(x)\chi_S(x)]$ is the corresponding inner product on Boolean functions, measuring intensity on a particular set of indices.

Recapping Fourier analysis

Definition ($\mathbb{F}_2$ polynomial)

$f : \mathbb{F}_2^n \to \mathbb{F}_2$ is a *polynomial of degree d* if it admits form $\sum_{S \subset \{1,\ldots,n\}} a_S \prod_{i \in S} x_i$, where the largest S for which a_S is nonzero has size $|S| = d$.

Definition (Fourier Growth)

The *Fourier growth* at the k-th level is

$$\mathcal{L}_{1,k}(f) := \sum_{S:|S|=k} |\hat{f}(S)| = \sum_{S:|S|=k} |\mathbb{E}_{x \sim \{-1,1\}^n}[f(x)\chi_S(x)]|$$

Why do we care?

- W.t.s. *P versus NP* (whether problems whose solutions are quickly verified can be quickly solved) but lack machinery to do so
    - Thus look at simpler objects
    - This includes $AC^0[2]$ *circuits*, which correspond to low-degree $\mathbb{F}_2$ polynomials
- Bounding $\mathcal{L}_{1,k}$ implies properties of $\mathrm{AC}^0[2]$ circuits! (Notably, may exist a *psuedorandom generator* fooling circuits of suitable size and depth)

Trivial bound

Lemma (Trivial bound on Fourier growth)

For f a polynomial on n coordinates, $\mathcal{L}_{1,k}(f) \leq \binom{n}{k}$

Proof:

- $\hat{f}(S) = \mathbb{E}_{x \in \{-1,1\}^n}[f(x)\chi_S(x)] \in [-1, 1]$ for all S
- There are $\binom{n}{k}$ possible $S : |S| = k$
- Thus, $\mathcal{L}_{1,k}(f) = \sum_{S:|S|=k} |\hat{f}(S)| \leq \binom{n}{k}$

Conjectures and prior results

Conjecture (E. Chattopadhyay, P. Hatami, S. Lovett, and A. Tal 2019)

$\mathcal{L}_{1,2}(Poly_{n,d}) = O(d^2)$

Theorem (E. Chattopadhyay, P. Hatami, S. Lovett, and A. Tal 2019)

$\mathcal{L}_{1,1}(\mathcal{F}) \leq t$ *implies* $\mathcal{L}_{1,2}(\mathcal{F}) \leq t \cdot O(\sqrt{n \log n})$

Theorem (E. Chattopadhyay, P. Hatami, K. Hosseini, S. Lovett 2018)

f with degree d has $\mathcal{L}_{1,k}(f) \leq (k2^{3d})^k$

Theorem (J. Błasiok, P. Ivanov, Y. Jin, C. H. Lee, R. A. Servedio, E. Viola 2024)

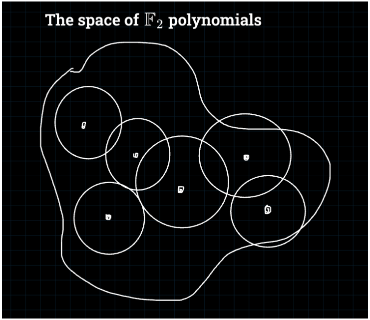
Satisfactory bounds exist for read-Δ, symmetric, and block-symmetric polynomials

Theorem (L. Becker, J. Slote, A. Volberg, H. Zhang 2024)

p with degree 2 has $\mathcal{L}_{1,k}(p) \leq (1+\sqrt{2})^k$

ϵ-net approach

- Find polynomial family $\mathcal{F}$ throughout the space of $\mathbb{F}_2$ polynomials with bounded $\mathcal{L}_{1,k}$
- Create an additivity rule so that some number of polynomials "around" $h \in \mathcal{F}$ have $\mathcal{L}_{1,k}$ bounded in terms of $\mathcal{L}_{1,k}(h)$



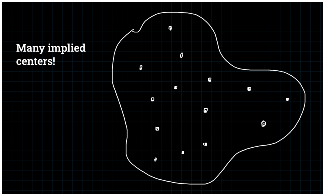
Our findings: Fourier growth for a random polynomial

- If we can show $\mathcal{L}_{1,k}(f)$ for f a random polynomial is *tightly concentrated* around a low expectation, then a supermajority of polynomials have bounded $\mathcal{L}_{1,k}$.

Theorem (Fourier growth for random polynomial)

Let $f = (-1)^{p(x)}$ for $p(x) = \sum_S B_S \prod_{i \in S} x_i$ and $B_S \sim$ i.i.d. Bern(1/2). Then for any m, t,

$$\Pr\left[\mathcal{L}_{1,k}(f) \geq t\right] \leq \frac{\binom{n}{k}^m 2^{-mn/2} (m-1)!!}{t^m}$$



Our findings: Fourier growth for a random polynomial

Theorem (Fourier growth for random degree d polynomial)

Let $f' = (-1)^{p'(x)}$ for $p'(x) = \sum_{S:|S|\leq d} B_S \prod_{i\in S} x_i$ and $B_S \sim$ i.i.d. Bern(1/2). Then

$$\mathbf{P}[\mathcal{L}_{1,k}(f') \geq t] \leq \frac{2^{-nm} \sum_{i\in\{1,\dots,m\}:|S_i|=k} \sum_{\forall i: r_i \in \{0,\dots,n\}} \left[\sum_{\forall i:|x^{(i)}|=r_i, \prod_{\forall i,j} x_i^{(j)}=1} 1 \cdot \prod_{j=1}^{m} |\mathcal{K}_k(r_j;n)|\right]}{t^m}$$

Lemma (I. Ben-Eliezer, R. Hod, S. Lovett 2013)

Fix $\epsilon > 0$ and let f be a random degree-d polynomial for $d \leq (1-\epsilon)n$. Then

$$\mathbf{P}[|\mathbb{E}[(-1)^{f(x)}]| > 2^{-c_1 n/d}] \leq 2^{-c_2\binom{n}{\leq d}}$$

for $c_1, c_2 \in (0,1)$ depending only on ϵ.

Our findings: additivity rule (kind of)

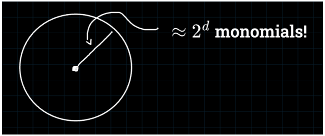
Theorem (Our current ϵ-ball, informally)

"Let h be from a function family $\mathcal{F}$ which is closed under restrictions and has bounded $\mathcal{L}_{1,k}(\mathcal{F})$. Then adding $\approx 2^d$ monomials of degree d to h does not blow up $\mathcal{L}_{1,k}$ too much."

Lemma (An additivity rule)

Let's say we have $h(x)$, r disjoint monomials $g_1(x), \ldots, g_r(x)$ each supported on T_i, so that altogether $rd = n$. We'll say $(-1)^{f_i(x)} := (-1)^{h(x)} \cdot \prod_{j=1}^{i} (-1)^{g_j(x)}$. We have

$$\mathcal{L}_{1,k}(f_r) \leq \mathcal{L}_{1,k}(h) + \frac{2}{2^d} k d^k \mathcal{L}_{1,k}(h|_{\textit{all 1s on } \bar{T}_1})$$



Our findings: additivity rule (kind of)

- Towards a general rule: if $g, h : \mathbb{F}_2^n \to \mathbb{F}_2$ have degree d,

$$\mathcal{L}_{1,k}(g+h) \approx \frac{1}{2^n} \sum_{x \in \{-1,1\}^n} (-1)^{g(x)} (-1)^{h(x)} \mathcal{K}_k(|x|, n)$$

Conclusion

Future stuff

- Find tighter degree-d concentration inequality
- Find more general additivity laws
- Explore approximations and properties of Kravchuk polynomials

Conclusion

Acknowledgments

- Vladimir Podolskii and Dale Jacobs
- Saeed Mehraban
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- DIAMONDS program

Extra: Recapping Fourier analysis

Definition (Fourier character)

The *Fourier character* for a set S is

$$\chi_S(x) = \prod_{i \in S} x_i$$

where $\chi_\emptyset = 1$ by convention.

Theorem (The Fourier expansion)

All functions $\mathbb{F}_2^n \to \mathbb{R}$ can be uniquely expressed as a multilinear polynomial

$$f(x) = \sum_{S \subset \{1,\ldots,n\}} \hat{f}(S)\chi_S(x)$$

called the Fourier expansion of f, where $\hat{f}(S)$ is the Fourier coefficient of f at S, and can be evaluated as $\hat{f}(S) = \mathbb{E}_{x \in \{-1,1\}^n}[f(x)\chi_S(x)]$.

Extra: the Kravchuk polynomial

Definition (Kravchuk polynomial)

A *Kravchuk* polynomial $\mathcal{K}_k$ is defined as the following:

$$\mathcal{K}_k(x; n, q) = \mathcal{K}_k(x) = \sum_{j=0}^{k} (-1)^j (q-1)^{k-j} \binom{x}{j} \binom{n-x}{k-j}$$

Definition (Alternate form for Kravchuk polynomial [O'Donnell 2014])

$\mathcal{K}_j(z; n) = \sum_{|S|=j} x^S$ for $\sum_i x_i = n - 2z$